

# Optimal Monetary Policy with Heterogeneous Agents: A Timeless Ramsey Approach

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Introduction

Model

Optimal monetary policy under discretion

Optimal monetary policy under commitment

Quantitative exercise – timeless policy

Conclusion

- What's the optimal monetary policy in a HANK model?
  - HANK vs. RANK
- Is the optimal policy time-consistent?
  - Commitment vs. Discretion
- Does Divine Coincidence still hold?
  - Response to different shocks

- Characterizes the optimal discretionary and Ramsey monetary policy in a continuous-time HANK model.
  1. Even without markup distortion, optimal Ramsey policy is **not** time-consistent.
  2. Discretionary policy features **inflationary bias** in the steady state
  3. Divine coincidence **does not** hold: short-run inflationary bias due to redistribution motive
- Adapts the sequence-space method to compute the optimal policy in a calibrated model.
  - Sequence-space perturbation around the stationary Ramsey plan
  - New: algorithm for computing the sequence-space Hessian for the dual Ramsey problem

# Roadmap

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# Model – Overview

- Continuous-time HANK model without aggregate uncertainty
  - Optimal policy in response to MIT shock
- Sticky wage + labor union with **uniform** rationing rule (a la Auclert et al.)
  - Every household supplies the same amount of labor  $\Rightarrow$  MRS not equalized across households
- No capital. One-period nominal bond in zero net supply.
- Neutral fiscal policy + zero government debt
  - Only affects the steady-state markup

- Utility

$$\mathbb{E}_0 \int_0^\infty e^{-\int_0^t \rho_s ds} \left[ u(c_t) - \left( v \left( \int_0^1 n_{k,t} dk \right) + \frac{\delta}{2} \int_0^1 (\pi_{k,t}^w)^2 dk \right) \right] dt$$

- Budget constraint

$$\begin{aligned} \dot{a}_t &= r_t a_t + z_t \frac{1}{P_t} \int_0^1 W_{k,t} n_{k,t} dk + \tau_t(z_t) - c_t \\ &= r_t a_t + \textcolor{brown}{z}_t \textcolor{brown}{w}_t \textcolor{brown}{n}_t - c_t \end{aligned} \quad (\text{in equilibrium})$$

- Borrowing constraint

$$a_t \geq \underline{a}$$

# Model – Labor union

- CES aggregator

$$N_t = \left( \int_0^1 N_{k,t}^{\frac{\epsilon_t-1}{\epsilon_t}} \right)^{\frac{\epsilon_t}{\epsilon_t-1}}, \quad N_{k,t} = \iint z n_{k,t}(a, z) g(a, z) da dz$$

- Union maximizes average utility of its members. detail Symmetric equilibrium + uniform rationing

$$N_t = N_{k,t} = n_{k,t}(a, z) \quad \forall (a, z)$$

- Wage PC

$$\begin{aligned} \dot{\pi}_t^w &= \rho_t \pi_t^w + \frac{\epsilon_t}{\delta} \left[ \frac{\epsilon_t - 1}{\epsilon_t} (1 + \tau^L) w_t \iint z u'(c_t(a, z)) g_t(a, z) da dz - v'(N_t) \right] N_t \\ &= \rho_t \pi_t^w + \frac{\epsilon_t}{\delta} \iint \tau_t(a, z) g_t(a, z) da dz \end{aligned}$$

# Model – firm and government

- Competitive final good market

$$Y_t = A_t N_t \quad \Rightarrow \quad \frac{W_t}{P_t} \equiv w_t = A_t$$

- Balance budget

$$P_t \tau_t(z) = \int_0^1 \tau^L z W_{k,t} n_{k,t} dk - \tau^L z W_t N_t = 0$$

- Inflation and Fisher equation

$$\pi_t = \pi_t^w - \frac{\dot{A}_t}{A_t}$$

$$r_t = i_t - \pi_t$$

# Equilibrium definition

Given an initial household distribution  $g_0(a, z)$ , a symmetric nominal wage distribution  $W_{k,0} = W_0$ , and given predetermined sequence of monetary policy  $\{i_t\}$ , shocks  $\{A_t, \rho_t, \epsilon_t\}$ , a *competitive equilibrium* is defined as paths for prices  $\{\pi_t^w, \pi_t, w_t, r_t\}$ , aggregates  $\{Y_t, N_t, C_t, B_t\}$ , individual allocation rule  $\{c_t(a, z)\}$ , and for the joint distribution over household bond holdings and idiosyncratic labor productivities  $\{g_t(a, z)\}$ , such that households optimize, unions optimize, final good producers optimize, and markets clear, that is,

$$Y_t = C_t = \iint c_t(a, z) g_t(a, z) da dz$$
$$0 = B_t = \iint a g_t(a, z) da dz$$

# Equilibrium conditions

## Micro block

$$\rho_t V_t(a, z) = u(c_t(a, z)) - v(N_t) - \frac{\delta}{2}(\pi_t^w)^2 + \partial_t V_t(a, z) + \mathcal{A}_t V_t(a, z) \quad (\text{HJB})$$

$$u'(c_t(a, z)) = \partial_a V_t(a, z) \quad (\text{individual FOC})$$

$$\partial_t g_t(a, z) = \mathcal{A}_t^* g_t(a, z) \quad (\text{FP})$$

## Macro block

$$\partial_t \pi_t^w = \rho_t \pi_t^w + \frac{\epsilon_t}{\delta} \iint \tau_t(a, z) g_t(a, z) da dz \quad (\text{Wage PC})$$

$$\iint c_t(a, z) g_t(a, z) da dz = \iint A_t z N_t g_t(a, z) da dz \quad (\text{Market clearing})$$

$\mathcal{A}_t$

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# Ramsey problem with finite commitment horizon

- Suppose a planner can commit to its policy for a finite horizon  $[0, \tau_1)$ .
- Taking as given the initial distribution  $g_0(a, z)$ , the shock  $\{A_t, \rho_t, \epsilon_t\}$ , and *future policy*, a planner chooses allocation and prices to maximize the utilitarian SWF, subject to implementability constraint (i.e. equilibrium conditions).
- Formally, a Ramsey problem with commitment horizon  $[0, \tau_1)$  solves

$$\mathcal{W}_0(g_0) = \min_{\mathbf{M}} \max_{\mathbf{X}} \mathbb{E}_0 \left[ L(0, \tau_1, g_0) + e^{-\int_0^{\tau_1} \rho_s ds} \mathcal{W}_{\tau_1}(g_{\tau_1}) \right]$$

$$\begin{aligned}
 L(t_1, t_2, g_{t_1}) := & \int_{t_1}^{t_2} e^{-\int_0^t \rho_s ds} \left\{ \iint \left\{ \mathcal{U}_t(a, z) g_t(a, z) \right. \right. \\
 & + \phi_t(a, z) \left[ -\rho_t V_t(a, z) + \mathcal{U}_t(a, z) + \partial_t V_t(a, z) + \mathcal{A}_t V_t(a, z) \right] \\
 & + \chi_t(a, z) \left[ u'(c_t(a, z)) - \partial_a V_t(a, z) \right] \\
 & \left. \left. + \lambda_t(a, z) \left[ -\partial_t g_t(a, z) + \mathcal{A}_t^* g_t(a, z) \right] \right\} da dz \right. \\
 & + \mu_t \left[ \iint (c_t(a, z) - A_t z N_t) g_t(a, z) da dz \right] \\
 & \left. + \theta_t \left[ -\partial_t \pi_t^w + \rho_t \pi_t^w + \frac{\epsilon_t}{\delta} \iint \tau_t(a, z) g_t(a, z) da dz \right] \right\}
 \end{aligned}$$

where  $\mathcal{U}_t(a, z) := u(c_t(a, z)) - v(N_t) - \frac{\delta}{2} (\pi_t^w)^2$

# Interpretation of the multipliers

$$\phi_t(a, z) \left[ -\rho_t V_t(a, z) + \mathcal{U}_t(a, z) + \partial_t V_t(a, z) + \mathcal{A}_t V_t(a, z) \right]$$

$$\theta_t \left[ -\partial_t \pi_t^w + \rho_t \pi_t^w + \frac{\epsilon_t}{\delta} \iint \tau_t(a, z) g_t(a, z) da dz \right]$$

$$\chi_t(a, z) \left[ u'(c_t(a, z)) - \partial_a V_t(a, z) \right]$$

$$\lambda_t(a, z) \left[ -\partial_t g_t(a, z) + \mathcal{A}_t^* g_t(a, z) \right]$$

$$\mu_t \left[ \iint (c_t(a, z) - A_t z N_t) g_t(a, z) da dz \right]$$

- $\phi_t(a, z), \theta_t$ : promise keeping
- $\chi_t(a, z)$ : social value of relaxing individual optimal consumption decision
- $\lambda_t(a, z)$ : social lifetime value of household at  $(a, z)$
- $\mu_t$ : social value of increasing aggregate demand

# Ramsey problem under discretion

- Suppose  $\{\tau_t\}$  is a Poisson process with rate  $\psi$ .
- The Ramsey problem under discretion is the *limit* of the Ramsey problem with finite commitment horizon when  $\psi \rightarrow \infty$ .
- Essentially, it means that the multipliers associated with the forward-looking constraints are zero. i.e.  $\phi_t(a, z) = 0$  and  $\theta_t = 0$ 
  - Only able to control the allocation at an instant of time.
  - Take  $V_t(a, z), \pi_t^w$  as given. They are pinned down by the equilibrium of the dynamic games among "planners".

## Policy under discretion: optimality condition

**Proposition 1.** The FONC that characterizes optimal policy under discretion are

$$\begin{aligned}\rho_t \lambda_t(a, z) &= \mathcal{U}_t(a, z) + \mu_t(c_t(a, z) - A_t z N_t) + \partial_t \lambda_t(a, z) + \mathcal{A}_t \lambda_t(a, z) \\ u'(c_t(a, z)) &= \partial_a \lambda_t(a, z) - \mu_t + \tilde{\chi}_t(a, z) \\ 0 &= \iint \left[ z \partial_a \lambda_t(a, z) g_t(a, z) + \underline{z} \zeta_t^{HTM} g_t(\underline{a}, \underline{z}) \right] da dz - \mu_t - \frac{v'(N_t)}{A_t} \\ 0 &= \iint \left[ a \partial_a \lambda_t(a, z) g_t(a, z) + \underline{a} \zeta_t^{HTM} g_t(\underline{a}, \underline{z}) \right] da dz\end{aligned}$$

where

$$\zeta_t^{HTM} \equiv u'(c_t(\underline{a}, \underline{z})) - \partial_a \lambda_t(\underline{a}, \underline{z}) + \mu_t, \quad \tilde{\chi}_t(a, z) \equiv -u''(c_t(a, z)) \frac{\chi_t(a, z)}{g_t(a, z)}$$

# Interpretation

- Social lifetime value takes into account effect on aggregate demand

$$\rho_t \lambda_t(a, z) = \mathcal{U}_t(a, z) + \mu_t(c_t(a, z) - A_t z N_t) + \partial_t \lambda_t(a, z) + \mathcal{A}_t \lambda_t(a, z)$$

$$\partial_a \lambda_t(a, z) = u'(c_t(a, z)) + \mu_t - \tilde{\chi}_t(a, z)$$

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$$\partial_a \lambda_t(a, z) = u'(c_t(a, z)) + \mu_t - \tilde{\chi}_t(a, z)$$

- FONC wrt.  $N_t$

$$\underbrace{\iint [\textcolor{brown}{z} \partial_a \lambda_t(a, z) g_t(a, z) + \underline{z} \zeta_t^{HTM} g_t(\underline{a}, \underline{z})] da dz}_{\text{social benefit from increasing } N_t} = \underbrace{\mu_t + \frac{v'(N_t)}{A_t}}_{\text{social cost}}$$

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$$\rho_t \lambda_t(a, z) = \mathcal{U}_t(a, z) + \mu_t(c_t(a, z) - A_t z N_t) + \partial_t \lambda_t(a, z) + \mathcal{A}_t \lambda_t(a, z)$$

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- FONC wrt.  $N_t$

$$\underbrace{\iint [z \partial_a \lambda_t(a, z) g_t(a, z) + \underline{z} \zeta_t^{HTM} g_t(\underline{a}, \underline{z})] da dz}_{\text{social benefit from increasing } N_t} = \underbrace{\mu_t + \frac{v'(N_t)}{A_t}}_{\text{social cost}}$$

- FONC wrt. nominal rate  $i_t$

$$\iint [\textcolor{brown}{a} \partial_a \lambda_t(a, z) g_t(a, z) + \underline{a} \zeta_t^{HTM} g_t(\underline{a}, \underline{z})] da dz = 0$$

# Targeting rule under discretion

**Proposition 2.** Optimal monetary policy under discretion is characterized by the targeting rule

$$\underbrace{\iint \left( zu'(c_t(a, z)) - \frac{v'(N_t)}{A_t} \right) g_t(a, z) da dz}_{\text{Aggregate Labor Wedge}} = \Omega_t^D \underbrace{\iint au'(c_t(a, z)) g_t(a, z) da dz}_{\text{Distributive Pecuniary Effect} < 0}$$

where

$$\Omega_t^D = \frac{\iint a(1 - \mathcal{M}_t(a, z)) g_t(a, z) da dz - (1 - \mathcal{M}_t(\underline{a}, \underline{z})) \underline{a} g_t(\underline{a}, \underline{z})}{\iint z(1 - \mathcal{M}_t(a, z)) g_t(a, z) da dz - (1 - \mathcal{M}_t(\underline{a}, \underline{z})) \underline{z} g_t(\underline{a}, \underline{z})} > 0$$

$\mathcal{M}_t(a, z)$

*Implication.* Discretionary policy tends to overheat the economy.

Output gap

# Inflationary bias under discretion

**Proposition 3.** The Markov perfect stationary equilibrium with optimal monetary policy under discretion features inflationary bias, given by

$$\pi_{ss}^w = \frac{\epsilon}{\delta} A_{ss} N_{ss} \left[ \underbrace{\left( 1 - \frac{\epsilon - 1}{\epsilon} (1 + \tau^L) \right) \Lambda_{ss}}_{\text{Monopolistic competition: } \geq 0} - \Omega_{ss}^D \underbrace{\text{Cov}_{g_{ss}(a,z)}(a, u'(c_{ss}(a, z)))}_{\text{Redistribution: } < 0} \right]$$

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*Corollary.* Even if  $\frac{\epsilon - 1}{\epsilon} (1 + \tau^L) = 1$ , steady state still features inflationary bias under discretion.

*Remark.* Quantitatively, the redistribution term is more important.

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# Ramsey problem under commitment

- The standard primal Ramsey problem corresponds to the limit  $\psi \rightarrow 0$ :

$$\min_{\mathbf{M}} \max_{\mathbf{X}} L^{SP}(g_0)$$

where

$$L^{SP}(g_0) := \lim_{T \rightarrow \infty} L(0, T, g_0)$$

## Policy under commitment: optimality condition (1/3)

**Proposition 4.** The FONC that characterizes optimal policy under commitment are

$$\rho_t \lambda_t(a, z) = \mathcal{U}_t(a, z) + \mu_t(c_t(a, z) - A_t z N_t) + \theta_t \frac{\epsilon_t}{\delta} \tau_t(a, z) + \partial_t \lambda_t(a, z) + \mathcal{A}_t \lambda_t(a, z)$$

$$u'(c_t(a, z)) = \partial_a \lambda_t(a, z) - \mu_t - \theta_t \frac{\epsilon_t}{\delta} \frac{d\tau_t(a, z)}{dc_t(a, z)} + \tilde{\chi}_t(a, z)$$

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$$u'(c_t(a, z)) = \partial_a \lambda_t(a, z) - \mu_t - \theta_t \frac{\epsilon_t}{\delta} \frac{d\tau_t(a, z)}{dc_t(a, z)} + \tilde{\chi}_t(a, z)$$

- Social lifetime value now also takes into account *individual* effect on inflation
  - $\theta_t$  is multiplier associated with wage PC,  $\tau_t(a, z)$  is individual labor wedge

## Policy under commitment: optimality condition (2/3)

### Proposition 4 (cont.)

$$\begin{aligned} 0 &= \iint [z \partial_a \lambda_t(a, z) g_t(a, z) + \underline{z} \zeta_t^{HTM} g_t(\underline{a}, \underline{z})] da dz - \mu_t - \frac{v'(N_t)}{A_t} \\ &\quad + \iint \phi_t(a, z) \left( z u'(c_t(a, z)) - \frac{v'(N_t)}{A_t} \right) da dz + \theta_t \frac{\epsilon_t}{\delta} \iint \frac{1}{A_t} \frac{d\tau_t(a, z)}{dN_t} g_t(a, z) da dz \\ 0 &= \iint [a \partial_a \lambda_t(a, z) g_t(a, z) + a \phi_t(a, z) \partial_a V_t(a, z) + \underline{a} \zeta_t^{HTM} g_t(\underline{a}, \underline{z})] da dz \end{aligned}$$

where  $\zeta_t^{HTM} \equiv u'(c_t(\underline{a}, \underline{z})) - \partial_a \lambda_t(\underline{a}, \underline{z}) + \mu_t + \theta_t \frac{\epsilon_t}{\delta} \frac{d\tau_t(\underline{a}, \underline{z})}{dc_t(\underline{a}, \underline{z})}$  and  $\tilde{\chi}_t(a, z) \equiv -u''(c_t(a, z)) \frac{\chi_t(a, z)}{g_t(a, z)}$ .

## Policy under commitment: optimality condition (2/3)

### Proposition 4 (cont.)

$$\begin{aligned} 0 &= \iint [z \partial_a \lambda_t(a, z) g_t(a, z) + \underline{z} \zeta_t^{HTM} g_t(\underline{a}, \underline{z})] da dz - \mu_t - \frac{v'(N_t)}{A_t} \\ &\quad + \iint \phi_t(a, z) \left( z u'(c_t(a, z)) - \frac{v'(N_t)}{A_t} \right) da dz + \theta_t \frac{\epsilon_t}{\delta} \iint \frac{1}{A_t} \frac{d\tau_t(a, z)}{dN_t} g_t(a, z) da dz \\ 0 &= \iint [a \partial_a \lambda_t(a, z) g_t(a, z) + a \phi_t(a, z) \partial_a V_t(a, z) + \underline{a} \zeta_t^{HTM} g_t(\underline{a}, \underline{z})] da dz \end{aligned}$$

where  $\zeta_t^{HTM} \equiv u'(c_t(\underline{a}, \underline{z})) - \partial_a \lambda_t(\underline{a}, \underline{z}) + \mu_t + \theta_t \frac{\epsilon_t}{\delta} \frac{d\tau_t(\underline{a}, \underline{z})}{dc_t(\underline{a}, \underline{z})}$  and  $\tilde{\chi}_t(a, z) \equiv -u''(c_t(a, z)) \frac{\chi_t(a, z)}{g_t(a, z)}$ .

- Take into account effect on inflation and *private value*.

## Policy under commitment: optimality condition (3/3)

### Proposition 4 (cont.)

$$\partial_t \phi_t(a, z) = -\mathcal{A}_t^* \phi_t(a, z) + \partial_a \chi_t(a, z)$$

$$\begin{aligned} \dot{\theta}_t = & \delta \pi_t^w \left( 1 + \iint \phi_t(a, z) da dz \right) \\ & + \underbrace{\iint \left[ a \partial_a \lambda_t(a, z) g_t(a, z) + a \phi_t(a, z) \partial_a V_t(a, z) + \underline{a} \zeta_t^{HTM} g_t(\underline{a}, \underline{z}) \right] da dz}_{=0} \end{aligned}$$

and  $\phi_0(a, z) = \theta_0 = 0$ .

## Policy under commitment: optimality condition (3/3)

### Proposition 4 (cont.)

$$\begin{aligned}\partial_t \phi_t(a, z) &= -\mathcal{A}_t^* \phi_t(a, z) + \partial_a \chi_t(a, z) \\ \dot{\theta}_t &= \delta \pi_t^w \left( 1 + \underbrace{\iint \phi_t(a, z) da dz}_{= 0 \text{ in ss}} \right) \\ &\quad + \underbrace{\iint \left[ a \partial_a \lambda_t(a, z) g_t(a, z) + a \phi_t(a, z) \partial_a V_t(a, z) + \underline{a} \zeta_t^{HTM} g_t(\underline{a}, \underline{z}) \right] da dz}_{= 0}\end{aligned}$$

and  $\phi_0(a, z) = \theta_0 = 0$ .

- **Proposition 5.** Optimal long-run inflation under commitment is **zero**.

## Policy under commitment: optimality condition (3/3)

### Proposition 4 (cont.)

$$\partial_t \phi_t(a, z) = -\mathcal{A}_t^* \phi_t(a, z) + \partial_a \chi_t(a, z)$$

$$\begin{aligned} \dot{\theta}_t &= \delta \pi_t^w \left( 1 + \iint \phi_t(a, z) da dz \right) \\ &\quad + \underbrace{\iint \left[ a \partial_a \lambda_t(a, z) g_t(a, z) + a \phi_t(a, z) \partial_a V_t(a, z) + \underline{a} \zeta_t^{HTM} g_t(\underline{a}, \underline{z}) \right] da dz}_{=0} \end{aligned}$$

and  $\phi_0(a, z) = \theta_0 = 0$ .

- **Proposition 5.** Optimal long-run inflation under commitment is zero.
- If implements zero inflation, then  $\theta_t = \theta_0 = 0 \ \forall t$ .

# Targeting rule under commitment

**Proposition 10.** Assume  $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$  and  $v(n) = \frac{n^{1+\eta}}{1+\eta}$ . We summarize the optimal monetary stabilization policy with the targeting rule

$$Y_t = \tilde{Y}_t^{RA} \left\{ \frac{\frac{\epsilon_t - 1}{\epsilon_t} (1 + \tau^L) \Omega_t^1 + \theta_t (1 - \gamma) \frac{\epsilon_t}{\delta} \Omega_t^2}{1 + \theta_t (1 + \eta) \frac{\epsilon_t}{\delta}} \right\}^{\frac{1}{\gamma + \eta}}$$

where  $\tilde{Y}_t^{RA}$  is the natural output in the RANK limit and

$$\Omega_t^1 = 1 + \frac{\iint zu'(c_t) \phi_t da dz}{\iint zu'(c_t) g_t da dz} + \frac{\iint zu''(c_t) \phi_t da dz}{\iint zu'(c_t) g_t da dz}$$
$$\Omega_t^2 = \frac{1}{1 - \gamma} \left( 1 - \gamma \frac{u'(C_t)}{u''(C_t)} \frac{\iint zu''(c_t) \phi_t da dz}{\iint zu'(c_t) g_t da dz} \right)$$

- In the RANK limit,  $\Omega_t^1 = \Omega_t^2 = 1$ . With appropriate subsidy  $\frac{\epsilon - 1}{\epsilon} (1 + \tau^L) = 1$ , we have Divine Coincidence (i.e.  $Y_t = \tilde{Y}_t^{RA}, \pi_t = 0$ ).
- Not true under HANK because  $\Omega_t^1, \Omega_t^2 \neq 1$ .

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# Time-inconsistency and timeless approach

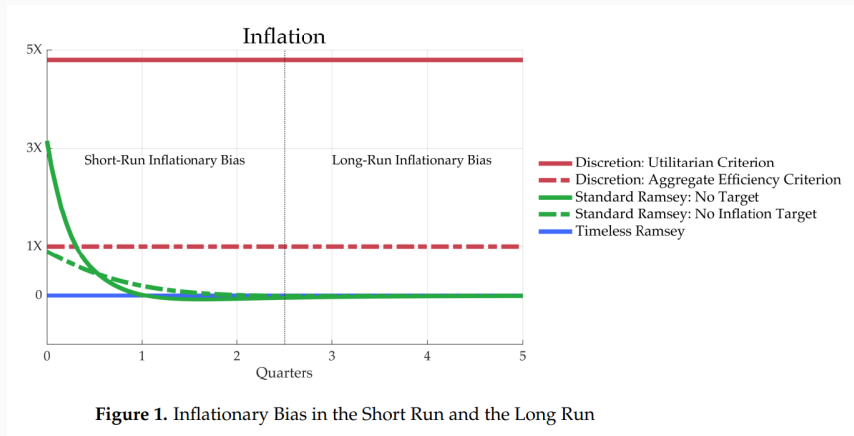
- Classic result: time-consistency requires  $\phi_{ss}(a, z) = \theta_{ss} = 0$ . (Marcet and Marimon 2019)
  - Does not hold in this model
- To ensure time-consistency, add to the objective function a *timeless penalty*

$$L^{TP}(g_0, \tau_{ss}, \theta_{ss}) := L^{SP}(g_0) + \iint \phi_{ss}(a, z) V_0(a, z) da dz - \theta_{ss} \pi_0^w$$

Essentially, the timeless penalty forces the initial condition  $\phi_0 = \phi_{ss}, \theta_0 = \theta_{ss}$  in the Ramsey FONC.

- Can interpret the timeless penalty as *inflation target* and *distribution target*.

# Inflationary bias under different policy



## Optimal timeless policy – perturbation

- We can summarize the Ramsey FONC and the equilibrium condition with a mapping in the *sequence space*

$$\mathcal{R} \left( \underbrace{\mathbf{X}}_{\text{allocation}}, \underbrace{\mathbf{M}}_{\text{multipliers}}, \underbrace{\mathbf{i}}_{\text{interest rate}}, \underbrace{\mathbf{Z}}_{\text{shocks}} \right) = 0$$

# Optimal timeless policy – perturbation

- We can summarize the Ramsey FONC and the equilibrium condition with a mapping in the *sequence space*

$$\mathcal{R} \left( \underbrace{\mathbf{X}}_{\text{allocation}}, \underbrace{\mathbf{M}}_{\text{multipliers}}, \underbrace{\mathbf{i}}_{\text{interest rate}}, \underbrace{\mathbf{Z}}_{\text{shocks}} \right) = 0$$

- Perturb the mapping around the *stationary* Ramsey plan

$$\mathcal{R}(\mathbf{X}_{ss}, \mathbf{M}_{ss}, \mathbf{i}_{ss}, \mathbf{Z}_{ss}) = 0$$

Note that this means  $\theta_0 = \theta_{ss}, \phi_0 = \phi_{ss}$

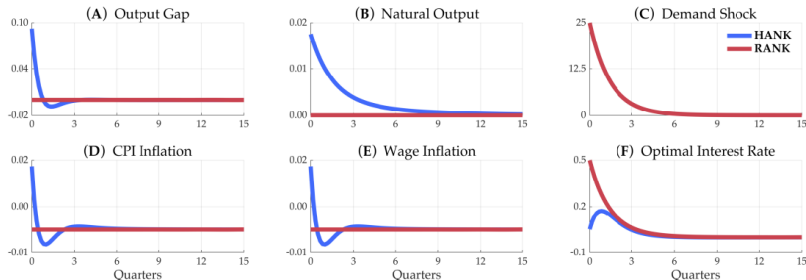
- This gives first-order approximation of the *timeless* Ramsey plan

$$d\mathbf{R} = -\mathcal{R}_{\mathbf{R}}^{-1} \mathcal{R}_{\mathbf{Z}} d\mathbf{Z}, \quad \text{where } \mathbf{R} := (\mathbf{X}, \mathbf{M}, \mathbf{i})$$

Can also use this Jacobian to solve for exact policy (quasi-Newton method).

- Preference:  $\rho = 0.02$ ,  $\gamma = \eta = 2$
- Union:  $\epsilon = 10$ ,  $\delta = 100 \Rightarrow$  slope of wage PC = 0.1
- Idiosyncratic productivity:  $z \sim$  two-state Markov chain,  $z^L = 0.8$ ,  $z^H = 1.2$ , transition rate = 0.33
- Employment subsidy:  $\frac{\epsilon-1}{\epsilon}(1 + \tau^L) = 1$

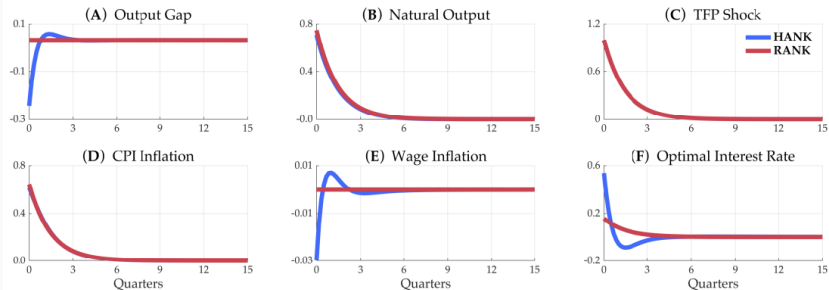
# IRF under timeless policy – Demand shock



**Figure 3.** Optimal Policy Transition Dynamics: Demand Shock

**Note.** Transition dynamics after a positive discount rate shock in both RANK (red) and HANK (blue) models under optimal monetary stabilization policy. Discount rate shock is initialized at  $\rho_0 = 0.025$  and mean-reverts to its steady state value  $\rho = 0.02$ , with a half-life of 1 quarter. Panels (A) through (C) report the dynamics of the output gap,  $\frac{Y_t - \bar{Y}_t}{\bar{Y}_t}$ , natural output, and the shock, all in percent deviations from the stationary Ramsey plan. Panels (D) through (F) report CPI inflation, wage inflation, and the optimal interest rate, all in percentage point deviations from the stationary Ramsey plan.

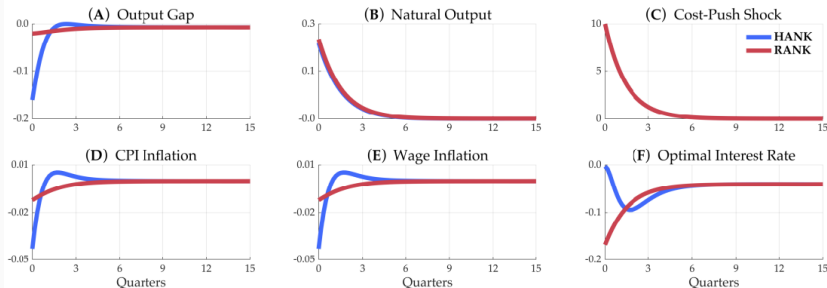
# IRF under timeless policy – TFP shock



**Figure 5.** Optimal Policy Transition Dynamics: TFP Shock

**Note.** Transition dynamics after a positive TFP shock in both RANK (red) and HANK (blue) models under optimal monetary stabilization policy. Initial shock is 1% of steady state TFP and mean-reverts with a half-life of 1 quarter. Panels (A) through (C) report the dynamics of the output gap,  $\frac{Y_t - \bar{Y}_t}{\bar{Y}_t}$ , natural output, and the shock, all in percent deviations from the stationary Ramsey plan. Panels (D) through (F) report CPI inflation, wage inflation, and the optimal interest rate, all in percentage point deviations from the stationary Ramsey plan.

# IRF under timeless policy – Cost-push shock



**Figure 6.** Optimal Policy Transition Dynamics: Cost-Push Shock

**Note.** Transition dynamics after a positive cost-push shock in both RANK (red) and HANK (blue) models under optimal monetary stabilization policy. The cost-push shock is modeled as an increase in labor union's desired wage mark-up. The shock is initialized at  $\epsilon_0 = 11$  and mean-reverts to its steady state value  $\epsilon = 10$ , with a half-life of 1 quarter. Panels (A) through (C) report the dynamics of the output gap,  $\frac{Y_t - \bar{Y}_t}{\bar{Y}_t}$ , natural output, and the shock, all in percent deviations from the stationary Ramsey plan. Panels (D) through (F) report CPI inflation, wage inflation, and the optimal interest rate, all in percentage point deviations from the stationary Ramsey plan.

# Roadmap

Introduction

Model

Optimal monetary policy under discretion

Optimal monetary policy under commitment

Quantitative exercise – timeless policy

Conclusion

- New source of time-inconsistency in HANK: **Redistribution motive**
  - Leads to inflationary bias both in the short-run and in the long-run (under discretion)
  - Breaks Divine Coincidence
- Augmenting inflation and redistribution target helps to ensure time-consistency

# Appendix

Union's objective

$$\max_{\pi_{k,t}^w} \int_0^\infty e^{-\int_0^t \rho_s ds} \left( \int \int \left[ u(c_t(a, z; W_{k,t})) - \left( v \left( \int_0^1 n_{k,t} dk \right) + \frac{\delta}{2} \int_0^1 (\pi_{k,t}^w)^2 dk \right) \right] g(a, z) dadz \right) dt$$

# Infinitesimal generator and adjoint

- The operator  $\mathcal{A}_t$  is given by

$$\mathcal{A}_t f(a, z) := (r_t a + w_t z N_t - c_t(a, z)) \partial_a f(a, z) + \mathcal{A}_z f(a, z)$$

- The operator  $\mathcal{A}_t^*$  is given by

$$\mathcal{A}_t^* f(a, z) := -\partial_a [(r_t a + z w_t N_t - c_t(a, z)) f(a, z)] + \mathcal{A}_z^* f(a, z)$$

- Importantly, we have

$$\int f(a, z) \mathcal{A}_t g(a, z) da dz = \int g(a, z) \mathcal{A}_t^* f(a, z) da dz \quad \forall f, g$$

When discretized,  $\mathcal{A}_t^*$  is the transpose of  $\mathcal{A}_t$ .

- The function  $\mathcal{M}_t(a, z)$  is given by

$$\mathcal{M}_t(a, z) = [\rho - (r_t - \partial_a c_t(a, z) + \partial_t + \mathcal{A}_t)]^{-1} \partial_a c_t(a, z)$$

- $r_t - \partial_a c_t(a, z)$ : saving
- $\partial_t$ : time adjustment
- $\mathcal{A}_t$ : risk adjustment

- Under discretion, we have

$$\partial_a \lambda_t(a, z) = \partial_a V_t(a, z) + \mathcal{M}_t(a, z) \mu_t$$

## Targeting rule under discretion – Output gap

- Assume  $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$  and  $v(n) = \frac{n^{1+\eta}}{1+\eta}$ . The targeting rule can be represented as

$$Y_t = \tilde{Y}_t \times \left( \frac{\epsilon_t}{\epsilon_t - 1} \frac{1}{1 + \tau^L} \right)^{\frac{1}{\gamma+\eta}} \times \underbrace{\left( 1 - \Omega_t^D \frac{\iint a u'(c_t(a, z)) g_t(a, z) da dz}{\iint z u'(c_t(a, z)) g_t(a, z) da dz} \right)^{\frac{1}{\gamma+\eta}}}_{>1}$$

where the natural output  $\tilde{Y}_t$  is given by

$$\tilde{Y}_t = \left( \frac{\epsilon_t - 1}{\epsilon_t} (1 + \tau^L) A_t^{1+\eta} \iint \frac{z u'(c_t(a, z))}{u'(C_t)} g_t(a, z) da dz \right)^{\frac{1}{\gamma+\eta}}$$

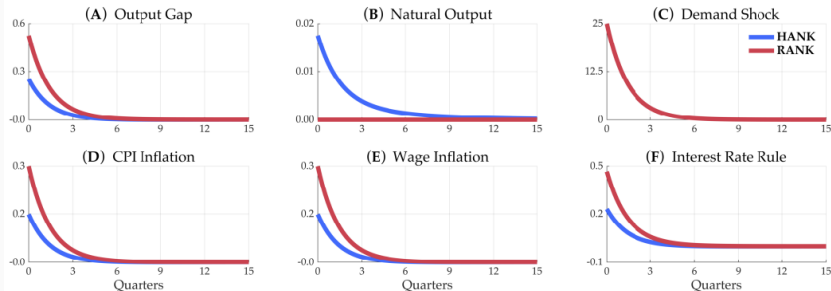
**Proposition 7.** With isoelastic (CRRA) preferences, the timeless penalty on inflation in both RANK and HANK economies is given by

$$\theta_{ss} := -\frac{\delta}{\epsilon} \frac{\Omega_{ss}^1 - (\tilde{Y}_{ss}^{RA})^{\gamma+\eta}}{\frac{\epsilon-1}{\epsilon}(1+\tau^L)(1-\gamma)\Omega_{ss}^2 - (1+\eta)(\tilde{Y}_{ss}^{RA})^{\gamma+\eta}}$$

where  $\tilde{Y}_{ss}^{RA} = \left[\frac{\epsilon-1}{\epsilon}(1+\tau^L)\right]^{\frac{1}{\gamma+\eta}}$

- Distributional considerations shape the inflation target

# IRF under Taylor rule – demand shock

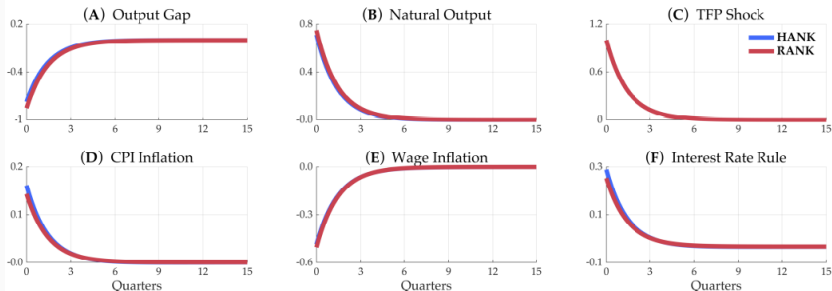


**Figure 10.** Transition Dynamics under Taylor Rule: Demand Shock

**Note.** Impulse responses to positive cost-push shock in both RANK (yellow) and HANK (blue) models. The nominal interest rate follows the Taylor rule (81) and is not set optimally. The cost-push shock is modeled as an increase in labor union's desired wage mark-up. The shock is initialized at  $\epsilon_0 = 11$  and mean-reverts to its steady state value  $\epsilon = 10$ , with a half-life of 2 quarters.

- Taylor rule:  $i_t = r_{ss} + 1.5\pi_t$

# IRF under Taylor rule – TFP shock

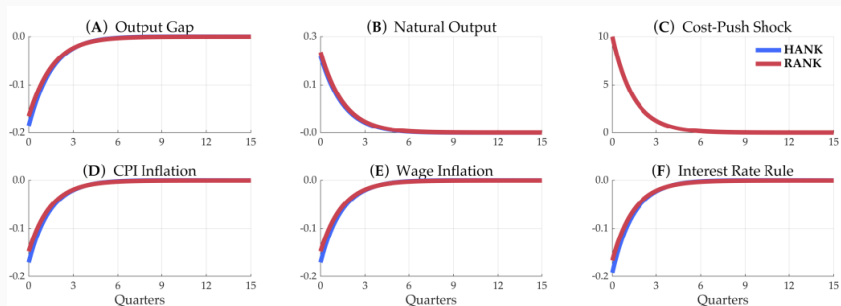


**Figure 9.** Transition Dynamics under Taylor Rule: TFP Shock

**Note.** Impulse responses to positive cost-push shock in both RANK (yellow) and HANK (blue) models. The nominal interest rate follows the Taylor rule (81) and is not set optimally. The cost-push shock is modeled as an increase in labor union's desired wage mark-up. The shock is initialized at  $\epsilon_0 = 11$  and mean-reverts to its steady state value  $\epsilon = 10$ , with a half-life of 2 quarters.

- Taylor rule:  $i_t = r_{ss} + 1.5\pi_t$

# IRF under Taylor rule – Cost-push shock

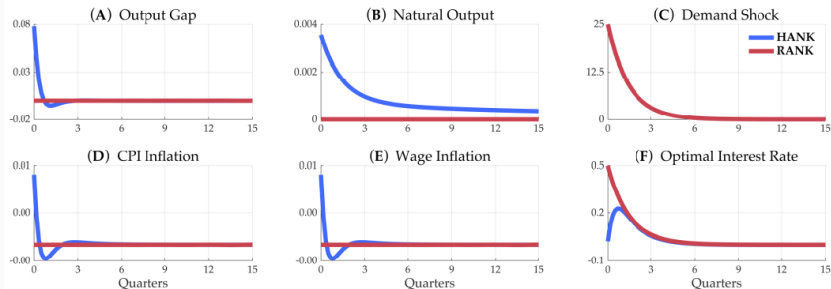


**Figure 11.** Transition Dynamics under Taylor Rule: Cost-Push Shock

**Note.** Impulse responses to positive cost-push shock in both RANK (yellow) and HANK (blue) models. The nominal interest rate follows the Taylor rule (81) and is not set optimally. The cost-push shock is modeled as an increase in labor union's desired wage mark-up. The shock is initialized at  $\epsilon_0 = 11$  and mean-reverts to its steady state value  $\epsilon = 10$ , with a half-life of 2 quarters.

- Taylor rule:  $i_t = r_{ss} + 1.5\pi_t$

# IRF under timeless policy – Demand shock, low earnings inequality

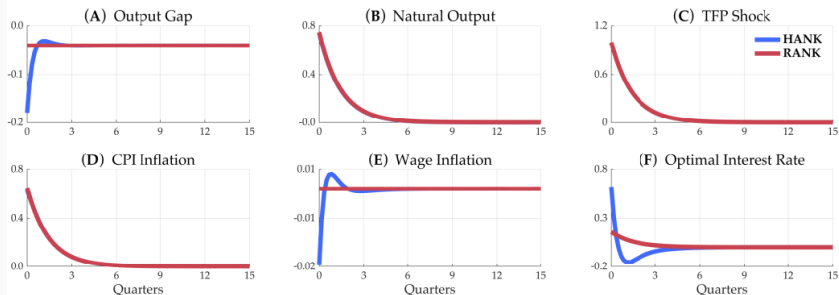


**Figure 13.** Optimal Policy Transition Dynamics with Low Earnings Risk: Demand Shock

**Note.** Impulse responses to positive discount rate shock in both RANK (yellow) and HANK (blue) models under optimal monetary stabilization policy, using an alternative calibration with less uninsurable earnings risk with  $z^L = 0.95$  and  $z^H = 1.05$ . The discount rate shock is initialized at  $\rho_0 = 0.025$  and mean-reverts to its steady state value  $\rho = 0.02$ , with a half-life of 2 quarters.

- Benchmark:  $z^L = 0.8, z^H = 1.2$

# IRF under timeless policy – TFP shock, low earnings inequality

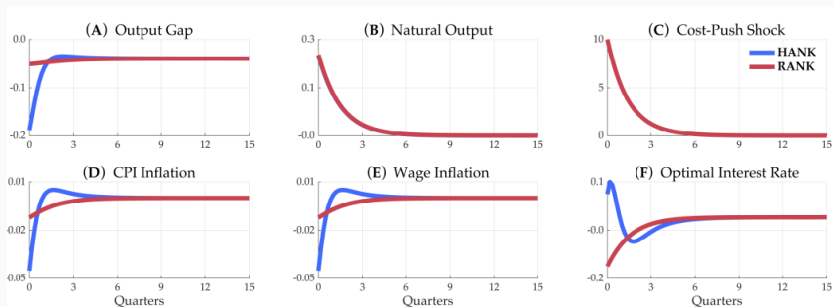


**Figure 12.** Optimal Policy Transition Dynamics with Low Earnings Risk: TFP Shock

**Note.** Impulse responses to positive TFP shock in both RANK (yellow) and HANK (blue) models under optimal monetary stabilization policy, using an alternative calibration with less uninsurable earnings risk with  $z^L = 0.95$  and  $z^H = 1.05$ . The initial shock is 1% of steady state TFP and mean-reverts with a half-life of 2 quarters.

- Benchmark:  $z^L = 0.8, z^H = 1.2$

# IRF under timeless policy – Cost-push shock, low earnings inequality



**Figure 14.** Optimal Policy Transition Dynamics with Low Earnings Risk: Cost-Push Shock

**Note.** Impulse responses to positive cost-push shock in both RANK (yellow) and HANK (blue) models under optimal monetary stabilization policy, using an alternative calibration with less uninsurable earnings risk with  $z^L = 0.95$  and  $z^H = 1.05$ . The cost-push shock is modeled as an increase in labor union's desired wage mark-up. The shock is initialized at  $\epsilon_0 = 11$  and mean-reverts to its steady state value  $\epsilon = 10$ , with a half-life of 2 quarters.

- Benchmark:  $z^L = 0.8, z^H = 1.2$